Six math clues · answer key

The drops that did not fall.

Age 7 / Grade 2 stretch. Offer tens-and-ones drawings, repeated addition, or oral reasoning.

Math 1 · 85

38 + 47 = 85. 8 + 7 = 15; 3 tens + 4 tens + 1 ten = 8 tens.

Math 2 · 34

72 − 38 = 34. Regroup as 6 tens and 12 ones: 12 − 8 = 4; 6 − 3 = 3.

Math 3 · 26

28 + 17 = 45; 45 − 19 = 26.

Math 4 · 24

6 + 6 + 6 + 6 = 24, so 4 × 6 = 24.

Math 5 · 33

27 + 3 = 30; 30 + 30 = 60. 3 + 30 = 33 more drops.

Math 6 · 27

22 + 5 = 27. The instruction adds 5 at each step.

Age 7 / Grade 2 stretch. Six tasks cover the learning contract. Offer counters, tens-and-ones sketches, or oral reasoning. Pause between pages; completing all six at once is optional.

Reading & writing · answer key

The drops that did not fall.

Reading · 1

The leaf and spider web were wet, but the path was dry and no rain had fallen. These clues made Little investigate another source of water.

Reading · 2

The leaf must become cool enough first. Air touching it cools; water vapor can then condense into drops. Cooling comes before noticing dew.

Reading · 3

The drops evaporated: liquid water changed back into invisible water vapor in the air.

Writing

Accept one sensible command sentence, spoken, dictated, or written. Example: “First cool the leaf, then wait for tiny drops.” The notebook describes nature; it does not control the weather.

The child is 7, in Grade 2. Follow his questions and welcome dictation. Dew forms on cool surfaces; it does not fall like rain. “Air holds vapor” is kid-sized language: the air is not a sponge. Cooling to the dew point allows vapor to condense. Clear skies favor radiative cooling; clouds and wind can reduce it, but do not guarantee a dry leaf. All math counts and diorama temperatures are pretend. No experiment is required. A cold glass is an analogy, not a required activity.

Accept the child’s spoken command, dictation, invented spelling, or writing. One clear sentence is enough. Example: “First cool the leaf, then wait for tiny drops.”

Four clues to keep

The drops that did not fall.

Quiz 1. What is water vapor?

Water vapor is invisible water in a gas form.

Quiz 2. What helps the leaf cool on a clear, calm night?

The leaf gives off heat to the open sky and can become colder than the surrounding air.

Quiz 3. What happens when air touching the leaf cools to its dew point?

Some vapor condenses into liquid dew right on the leaf. At the dew point, the air is cool enough for condensation.

Quiz 4. What happens as sunrise warms the dew?

Liquid dew evaporates into invisible water vapor. The water changes form; it is not gone forever.

The child is 7, in Grade 2. Follow his questions and welcome dictation. Dew forms on cool surfaces; it does not fall like rain. “Air holds vapor” is kid-sized language: the air is not a sponge. Cooling to the dew point allows vapor to condense. Clear skies favor radiative cooling; clouds and wind can reduce it, but do not guarantee a dry leaf. All math counts and diorama temperatures are pretend. No experiment is required. A cold glass is an analogy, not a required activity.